Wipro — Aptitude Test 2

Questions 26–50 of 100 · 25 questions

Question 26. What is vale of ‘a’ if a is integer provided that a4=1 and a3+1=0?

Solution:
a4 = 1
From this we can derive a = +1 or a = -1
If a = +1
Then a3 + 1 = 2
It does not satisfy.
If a = -1, then a3 + 1 = 0
So, a = -1.

Question 27. The time showed by an analog clock at a moment is 11 am, then 1234567890 hours later it will show time as?

Solution:
We have to find the reminder of 1234567890 when divided by 24.
Reminder of 1234567890/24 = 18
The time will be 18 hours past 11 am
= 5 am.

Question 28. Out of 52 students 35 can speak Hindi, 32 can speak English, 31 can speak German, 20 speak both Hindi and English, 18 speak both Hindi and German and 24 both English and German. How many can speak

all
languages?

Solution:
Solving using Venn Diagram.

Hindi
English

35
32

20

18
24
x

31

German

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Assume that the number of persons who speak all three languages = x
Number of persons who speak Hindi and English alone = 20 – x
Number of persons who speak Hindi and German alone = 18 – x
Number of persons who speak English and German alone = 24 – x
Hindi alone = 35 – (20 – x) – (18 – x)
English alone = 32 – (20 – x) – (24 – x)
German alone = 31 – (18 – x) – (24 – x)
Total
=
(20–x)+(18-x)+(24-x)+(35-(20-x)-(18-x))+(32-(20-x)-(24-x))+(31-(18-x)-(24-x))+x = 52
By solving this we, get,
44 + 4x = 52
x = 2. Two students speak all languages.

Question 29. A team of 36 members is divided into groups of equal size to make a trip. Since the groups were too large to fit in a car , 3 members were taken from each group and these members then formed into two additional groups . After this, all the groups had the same number of members. How many members of the team were in a group before the three

members were taken out from each group?

200 by adding like this:
2 + 4 + 6 + …
= 2(1+2+3+4+…)
= 2(1+2+3+…13)
= 2(91)
= 182 m
In 13 x 2 = 26 seconds, the monkey would have climbed 182 m.
In the 27th second the monkey will climb another 28 meters and by that it would have climbed more than the height of the building.
The monkey will reach the top in 27 seconds.

Solution:
Assume that there were N members before splitting the group.
After splitting there are N – 3 members per group.
Both N and N – 3 are factors of 36 because in both the cases there were equal number of members in each group.
The possible values for N and N -3 are (9 and 6) or (12 and 9).
If there were 12 members per group initially, there were 3 groups and after removing there will be 9 members who will form 1 group. This does not satisfy the condition.
If there were 9 members in a group before splitting, there were 4 groups from which 12 members will be removed. They will form 2 additional groups.
The number of members per group originally = 9.

Question 30. A monkey is climbing a 200 meter tall building. It starts from the ground and climbs 4m in 1st second, slips 2m in 2nd second, climbs 6m in 3rd second, then slips 2m in 4th second, climbs 8m in 5th

Solution:
The effective distance climbed by the frog in two minutes = 3 – 2 = 1 feet.
To climb 57 – 3 = 54 feet, the frog will take 54 x 2 = 108 minutes.

At 109th minute the frog will climb 3 feet and reach the top of the well.
Time taken = 109 minutes.

second, slips 2m in the next and so on. When would the monkey reach the top of the building?

Solution:
In the first two seconds the monkey climbs 2m. In the next two it climbs 4 and in the next two it climbs 6.
The height increases as a multiple of 2.
The nearest value we will get less than

Question 31. A man driving a car sees his speedometer and the number is a palindrome. The number is 13931 km. after driving for 2hr then he see the another palindrome number. find the speed of the car?

Solution:
The next palindrome number will be 14041.
The distance travelled between the two points = 110 km.
Time taken = 2 hours.
Speed = 110/2 = 55kmph.

Question 32. A frog can climb up a well at 3 feet per min but due to slipperiness of the well, frog slips down 2 feet before it starts climbing the next minute. If the depth of the well is 57 feet, how much time will the frog take to reach the top?

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Question 33. There are 27 balls, of which 1 is heavier. Given a

weight balance, minimum how many times you need to weigh to find out the odd ball?

Solution:
Split 27 balls into 3 groups of 9 balls each.
Weigh any two groups in the balance.
If both are of same weight, take the group which was kept outside for the measurement.
If one of the group goes down, take that group for the next measurement.
Out of the one group which got selected, split the 9 balls in 3 groups of three balls in each.
Repeat the above process to select a group of three balls in which one is overweight.
Out of the three balls selected, take two balls and place them on separate plates of the balances.
If they are of same weight, the ball kept outside is of more weight.
If one of them goes down, then that ball is overweight.
Total measurements required = 3
27 = 33. The power value of three will always be the answer.

Question 34. An ore contains 25% of an alloy that has 90% iron. Other than this, in remaining 75% of the ore there is no iron. How many kilogram of the ore is needed to obtain 60 kg. of pure iron?

Solution:
If there is a 100kg ore, there will be 25kg of alloy.
Quantity of iron in the 25 kg alloy
= (90/100)25 = 22.5kg.
For 22.5kg iron we need 100kg ore.
The quantity of ore required to get 60kg of iron:
(22.5/60) = (100/x)
x = 266.67.
The quantity of ore required is 266.67kg

Question 35. The average of 7 numbers is 50. The average of first three of them is 40, while the average of the last three is 60. What must be the remaining number?

Solution:

Sum of all the 7 numbers = 7 x 50 = 350
Sum of first three numbers
= 3 x 40 = 120
Sum of the last three numbers
= 3 x 60 = 180
Sum of first three and the last three numbers = 120+ 180 = 300
Value of the remaining number
= 350 – 300 = 50.

Question 36. The average age of a class of 39 students is 15 years. If the age of the teacher be included, then the average increases by 3 months. Find the age of the teacher.

Solution:
Average age increases by three months or 0.25 years after adding the teacher's age.
New average = 15.25
New total age = 15.25 x 40 = 610
Old average = 15
Old total age = 15 x 39 = 585.
Age of teacher = 610 – 585 = 25.

Question 37. A 10-litre mixture of milk and water contains 30 percent water. Two liters of this mixture is taken away. How many liters of water should now be added so that the amount of milk in the mixture is double that of water?

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Solution:
After removing 2 liters of the mixture, the remaining quantity will be
= 10 – 2 = 8 liters.
Quantity of water in the 8 liter mixture
= (30/100)8 = 2.4 liters.
Quantity of milk = 8 – 2.4 = 5.6 liters.
Amount of milk should be double that of water.
Amount of milk = 5.6.
So the amount of water should be
= 5.6/2 = 2.8 liters.
Available quantity of water
= 2.4 liters

Quantity of water to be added
= 2.8 – 2.4
= 0.4 liters = 400ml.

Quantity of tin in 16kg of 2nd alloy
= 16(3/4) = 12kg.
Total quantity of copper before adding additional copper
= 8 + 4 = 12kg.
Total quantity of Tin
= 2 + 2 = 14
The ratio of Copper and Tin at present
= 12 : 14
The required ratio
= 3 : 2 = 21 : 14
The quantity of Copper to be added
= 21 – 12 = 9kg.

Question 38. Gavaskar's average runs in the first 50 innings was 50. After the 51st innings his average became 51. How many runs did he score in the 51st innings?

Solution:
Total score of Gavaskar till the first fifty innings = 50 x 50 = 2500.
After 51st innings, his total score
= 51 x 51 = 2601
His score in the 51st innings
= 2601- 2500 = 101
He scored 101 runs in 51st match.

Question 39. The average of a set of numbers is 46. If 4 numbers whose average is 52 are subtracted from this set, the average becomes 44.5. Fin the original number of numbers in the set.

Solution:
Let us assume that there were N number of numbers initially.
Total of N numbers = N x 46 = 46N
After removing 4 numbers, the total of N-4 numbers = (N-4)44.5 = 44.5N – 178
Difference between the two sets
= 4 x Average of removed 4 numbers
= 4 x 52 = 208
So,
46N – 44.5N – 178 = 208
1.5N = 30
N = 20
There were 20 numbers initially.

Solution:
Let us assume that the capacity of the can is 100 liters.
Out of 100 liters 60 liters is filled with 1st

type. And 40 liters is filled with 2nd type.
Quantity of kerosene in 60 liters of first type = (25/100)60 = 15 liters.
Quantity of Kerosene in 40 liters od second type = (30/100)40 = 12 liters.
Total quantity of kerosene = 15 + 12 = 27 liters.
Percentage of kerosene = 27%.

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Question 40. In two alloys, copper and tin are related in the ratios of 4:1 and 1:3. 10kg of 1st alloy, 16 kg of 2nd alloy and some pure copper are melted together. An alloy is obtained in which the ratio of copper and tin was 3:2. Find the weight of the new alloy.

Solution:
The profit share is directly proportional to the amount invested and duration off investment.
P1 : P2 : P3 = T1I1: T2I2 : T3I3
P1 : P2 : P3 = 12x20 : 4x25 : 8x15
P1 : P2 : P3 = 240: 100 : 120 = 12 : 5 : 6
Share of Mukesh = (6/23)4600 = 1200.

Solution:
Quantity of copper in 10kg of 1st alloy = 10(4/5) = 8 kg
Quantity of tin in 10kg of 1st alloy = 10(1/5) = 2 kg
Quantity of copper in 16 kg of 2nd alloy = 16(1/4) = 4kg

Question 41. One type of liquid contains 25% of Kerosene, the other contains 30% of Kerosene. A can is filled with 6 parts of the first liquid and 4 parts of the second liquid. Find the percentage of the Kerosene in the new mixture.

Question 42. Rajan and Rakesh started a business and invested Rs. 20,000 and Rs. 25,000 respectively. After 4 months Rakesh left and Mukesh joined by investing Rs. 15,000. At the end of the year there was a profit of Rs. 4,600. What is the share of

Mukesh?

Question 43. A starts business with Rs.3500 and after 5 months, B joins with A as his partner. After a year, the profit is divided in the ratio 2 : 3. What is B's contribution in the Capital?

Solution:
Pa : Pb = T1T1 : T2I2
T1 = Duration of investment of A = 12
T2 = Duration of investment of B = 7
I1 = Investment amount of A = 3500
I2 = ?
Pa : Pb = 12 x 3500 : I2 x 7
2 : 3 = 6000 : I2
I2 = 9000
Investment of B = Rs. 9000

Question 44. Ratio between 2 numbers is 5:7 and their product is 560. What is the difference between 2 numbers?

Solution:
Assume that the two numbers are 5x and 7x.
Given that, (5x)(7x) = 560
35x2 = 560
x2 = 16 ■x = 4
The two numbers are, 5(4) = 20 and 7(4) = 28.
Difference between two numbers = 8

Question 45. A and B together have Rs. 2412. If 8/25 of A's amount is equal to 2/5 of B's amount then how much amount does A have?

Solution:
(8/25)A = (2/5)B
■(8/25)A = (10/25)B
So, 8A = 10B
A : B = 5 : 4
Share of A = (5/9)2412 = Rs. 1340.

Question 46. A shopkeeper keeps MRP of a product as Rs 45. But he sells it at Rs. 42.

Then after he gains a profit of 5%. What is the cost price of the product?

Solution:
Profit % = [(SP – CP)/CP]100

(105/100)CP = 42
CP = 42 x 100/105
Cost price of the product = Rs. 40.

Question 47. In a certain school, 20% of the students are below 8 years of age. The number of students above 8 years of age is (2/3) of the number of students of 8 years age which is 96. What is the total number of students in the school?

Solution:
Number of students whose age is exactly 8 years = 96.
Number of students above 8 years age = (2/3) x 96 = 64.
Total number of students with age 8 years and above = 96 + 64 =160.
Percentage of students below 8 years age = 20%
Therefore, Percentage of students with age 8 years or above = 100 – 20 = 80%
Let N be the total number of students in the school.
(80/100)N = 160
N = 200.
There are 200 students in the school.

Question 48. There are 5000 voters in a town out of which 20% are not eligible to vote and there are two candidates contesting. The winning candidate won by 15% of votes. What is the number of votes he got?

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Solution:
The number of persons eligible for voting = 5000 – (20/100)5000
= 4000 votes.
The winning candidate got 15% more votes than the losing candidate.
Let % of votes by winning candidate = X and % of votes by losing candidate = Y
X + Y = 100
X – Y = 15
By solving the above two equations, we get
X = 57.5
Percentage of votes got by winning

candidate = 57.5%
Votes he got = (57.5/100)4000 = 2300

Solution:
Let N be the number of years at 5%
simple interest.
The number of years for which the
amount is lent at 6% simple interest
= N – 2
In both the cases the amount got back is
same.
Amount = P + PNR/100
Amount in first case
= P + [P x (N – 2) x 6/100]
Amount in second case
= P + [P x N x 5/100]
Given that both amounts are equal.
So,
P+[Px(N-2)x6/100] = P+[PxNx5/100]
(N – 2) x 6 = N x 5
6N – 12 = 5N
N = 12 years.
The amount lent at 5% interest
= P + (P x 12 x 5/100) = 2800
= 1.6P = 2800
P = 1750.
The amount invested in each case
= Rs. 1750.

Question 49. A shopkeeper gives a discount of 20% on the sale. By what percent he had to increase the selling price of the item so that after giving discount he gets the cost price?

Solution:
Even after giving 20% discount the
shopkeeper does not incur loss or gain.
That means 80 percentage of selling
price = Cost price.
Assume that the cost price = Rs. 100
So, (80/100)S.P = 100
Selling price = Rs. 125
The shopkeeper must increase the price
from 100 to 125.
Percentage increase
= [(125-100)/100]100 = 25%
The cost price must be increased by 25%

Question 50. If the price of petrol increases by 25% and Kevin intends to spend only 15% more on petrol, by how much percent should he reduce the quantity of petrol that he buys?

Solution:
Assume that the price of 1 liter petrol = Rs. 100.
After increasing the price by 25%, the price of 1 liter petrol = Rs. 125.
But Kevin is ready to spend only 15% more on petrol.
The amount he is willing to spend = 100+(15/100)100 = Rs. 115
The price to be spent decreases fro 125 to 115.
Percentage decrease in quantity = [(125 – 115)/125]100 = 8%
Kevin reduces the petrol consumption by 8%